

PD 4

①) SEA $T=(V;E)$ CON $|V(T)|=6$
X LEMA DE APRETÓN DE MANOS.

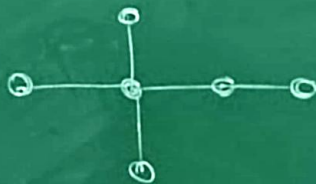
$$\sum_{i=1}^6 d_i = 2|E|$$

$|V|-1$ (X ÁRBOL)

EN EL PROB., $\sum_{i=1}^6 d_i = 10$

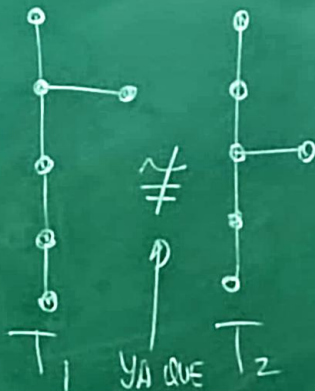
①^o $(1;1;1;1;1;5)$

②^o $(1;1;1;1;2;4)$



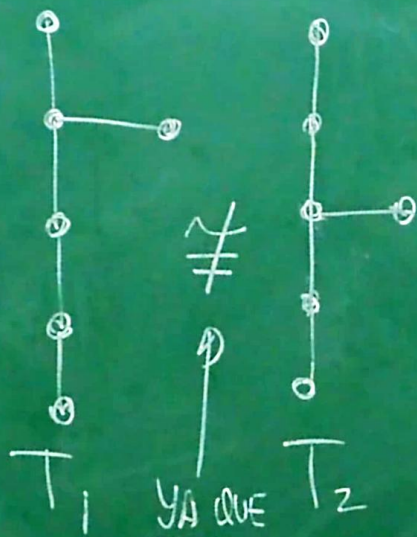
③^o $(1;1;1;1;3;3)$

④^o $(1;1;1;2;2;3)$



$\neq$
 $\uparrow$
YA QUE
 T_1 POSEE 2 SUBGRUPOS
ISOMORF. A P_5 Y T_2
NO LOS POSEE

⑤^o $(1;1;2;2;2;2)$



YA QUE
 T_1 POSEE 2 SUBGRAFOS
 ISOMORF. A P_5 Y T_2
 NO LOS POSEE

⑤ (1, 1, 2, 2, 2, 2)

② LEMA: SEA $T = (V, E)$ ÁRBOL
 CON $|V(T)| \geq 2$

→ T POSEE AL MENOS 2 HOJAS.

EN EL PROB.; SEA $T = (V, E)$ ÁRBOL
 COMO $\exists u \in V(T)$ CON $\deg(u) = k$

→ $T - u$ POSEE " k " COMPONENTES CONEXAS
 (BOSQUE)

SUPONG. QUE $T_1, T_2, \dots, T_k$ SON DICHAS
 COMPONENTES.

SI $(1^o) |V(T_p)| = 1, \forall p \in I_k$

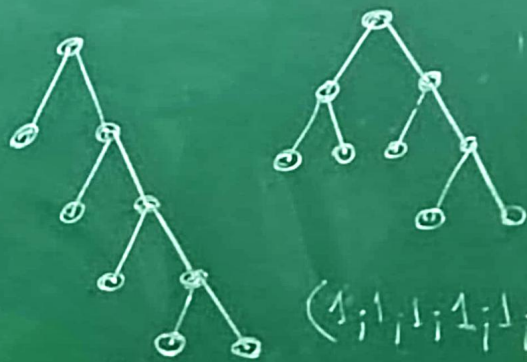
→ NO HAY NADA QUE PROBAR.

$(2^o) \exists j \in I_k / |V(T_j)| > 1$

x/ENA → T_j POSEE AL MENOS 2 HOJAS.

→ $\# \text{HOJAS}_{DET} > k$

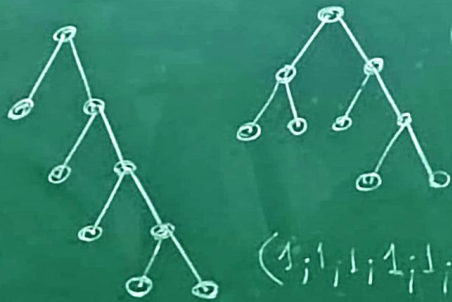
(4)



$(1, 1, 1, 1, 1, 2, 3, 3, 3)$

T_1 T_2
BINTREOS.
 $\sum_{i=1}^9 d_i = 2(8) = 16$

④


$$(1; 1, 1, 1, 1, 2, 3, 3, 3)$$
 T_1
$$T_2$$

BUNARLOS.

$$\sum_{i=1}^9 d_i = 2(8) = 16$$

$$\begin{aligned} f_2(f_2(a)) &= a \\ f_2(f_2(b)) &= b \\ f_2(f_2(c)) &= c \end{aligned}$$

5) SEA $T=(V;E)$ ÁRBOL
CON CÓDIGO. y $|V|=n$

$$(k_1; k_2; \dots; k_{2n})$$
$$\begin{array}{cc} \downarrow & \downarrow \\ 2 & 2 \end{array}$$

五二

$\rightarrow \# \text{ ARBORES} \leq 2^{2^n} = 4^n$

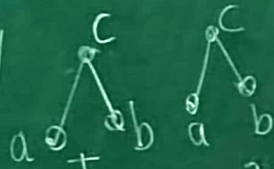
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$$\text{Aut}(T) = \langle \text{id}, f_2 \rangle$$

⑥

⑥ EL CONT. DE AUTOMORF.

$$\text{Aut}(G) = \{ f : V(G) \rightarrow V(G) \mid f \text{ IS ISOMORF. DE GRAFOS} \}$$

$$E \cap M \neq \emptyset \quad S \in A$$


if 1 $\varphi_d = f_1: V(T) \rightarrow V(T)$
 $a \mapsto a$
 $b \mapsto b$
 $c \mapsto c$

$$f_2: V(I) \rightarrow V(I)$$

$$\begin{array}{lcl} a & \mapsto & b \\ b & \mapsto & a \\ c & \mapsto & c \end{array}$$

DEF: UN GRAFO $G=(V;E)$
ES SIMÉTRICO.

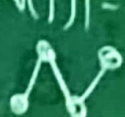
SI $|Aut(G)| > 1$

PE) UN GRAFO $G=(V;E)$
ES ASIMÉTRICO

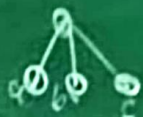
SI $|Aut(G)| = 1$

a) EN EL PROBLEMA
SEA $T=(V;E)$ ÁRBOL

1º) $|V(T)| = 4$

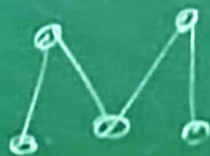


(1; 2; 2)



(1; 1; 1; 3)

2º) $|V(T)| = 5$

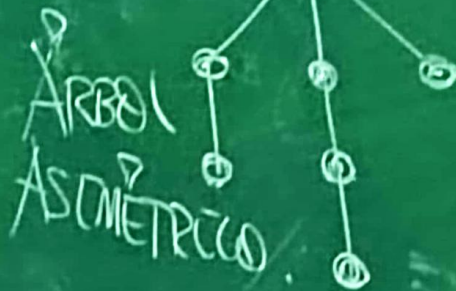


(1; 1; 2; 2; 2)



(1; 1; 1; 1; 4)

3º) $|V(T)| = 6$ (VERIFICAR)



ÁRBOL
ASIMÉTRICO

